

Daniel Oppenheimer

hyv6fx@virginia.edu | github.com/DannyOppenheimer | linkedin.com/in/daniel-oppenheimer | dannypoppenheimer.com

Education

- **University of Virginia** **Aug 2023 – May 2027**
- BS in Computer Science, Data Science Minor, Engineering Business Minor
- Cumulative GPA: 3.78
- Relevant Coursework: Cloud Computing, Parallel Computing, Machine Learning, Artificial Intelligence, Computer Networks, Natural Language Processing (NLP), Software Development Essentials, Software Engineering, Data Structures and Algorithms I & II, Computer Systems and Organization I & II, Discrete Mathematics I & II, New Product Development, Data Science Systems
- Activities and Societies: Trigon Engineering Society @ UVA, Sigma Alpha Mu – Special Events Chair, 2025 Deloitte National Undergraduate Case Competition, Intramural Sports Supervisor

Experience

Comcast, Software Engineer Intern

June 2026 – August 2026

- Assisted migration of Xfinity's customer-facing website's CI/CD pipeline from Concourse to GitHub Actions, eliminating hour-long delays per PR & streamlined releases for a service for 30 million customers.
- Cleared backlog tickets improving technical infrastructure for a team of ~50 by batching API logs/reducing console clutter and built tools to update stale development routes/allow for production payment flow testing.
- Architected new Wi-Fi troubleshooting menu for Xfinity app to drastically reduce frivolous customer support contacts and customer churn rate - pitched to senior leadership.

Booz Allen Hamilton, AI Software Engineer Intern

June 2025 – August 2025

- Engineered and benchmarked three distinct Retrieval-Augmented Generation (RAG) frameworks (Keyword, Hybrid, and Graph RAG) for an AI search tool, directly collaborating with NOAA developers to choose the optimal framework to streamline information retrieval from a complex data storage system.
- Implemented a novel hybrid search system integrating BM25 keyword ranking with semantic search, leveraging a Reciprocal Rank Fusion pipeline and a genetic algorithm to optimize search result relevancy.
- Developed a full-stack solution with a Flask frontend, connecting to a backend on AWS EC2. Used REST API strategies to connect to NOAA data, a Redis cache, and the Gemma3 GenAI model.
- Developed CoT system to auto-upload new data formats in a metadata-rich format.

Zbooni, Software Development Consultant

Dec 2025 – Jan 2026

- Engineered a centralized fraud detection dashboard to automate risk assessment for the compliance team, replacing manual oversight with robust security rules, saving KYC team members 3 hours per workday.
- Architected an automated email verification workflow using Trengo for suspicious transactions, streamlining customer confirmation responses and significantly reducing the platform's exposure to fraudulent activity.
- Prototyped Random Forest and XGBoost models for risk data collected through platform in 6 months.

Glass House Recycling LLC, Co-founder

Feb 2020 – August 2025

- Founded curbside glass-pickup business at age 14 to meet demand after Falls Church City stopped providing municipal glass recycling service.
- Developed in-house invoicing system with a Node.js express web application.
- Leveraged digital advertising and print/radio media (NBC, WAMU, NowThis) to grow customer base to over 160 residential households, online stories about business reached 550k+ views, with 5k+ reactions.
- \$10k+ exit, selling company and converting over 80% of customers to the new platform.

Projects

Drawtex, Developer

Dec 2025 – Present

- Designed AI driven hand-drawn notes app that classifies pen strokes between text, math, and diagrams using a Bi-Directional LSTM model along with advanced feature engineering to achieve an F1 score of 99%.
- Developed advanced converter that formats classified text into Latex, enhancing students' learning workflow.

Bracket Hub, Developer

Apr 2024 – Nov 2025

- Built social/sporting web application to bring a March Madness point-scoring system to user-made tournament brackets.
- React.js that interfaced with Firebase authentication and Firestore database for backend and data storage.

Spyfall, Developer

Jan 2020 – Apr 2020

- Designed minimalist reimagination of the "Spyfall" game front-end webpage using Node.js/HTML/CSS.
- Implemented WebSocket communication using the Socket.io API for live multiplayer capability.

Skills

- Python, JavaScript (React, Node.js, Typescript), Java, HTML/CSS, SQL
- Git, PyTorch, RAG, Linux, GitHub Actions, Concourse, Docker, Ollama, Firebase, Flask, Django, REST API, MATLAB, Agile Method, Jupyter Notebook, Blender

Awards: Eagle Scout